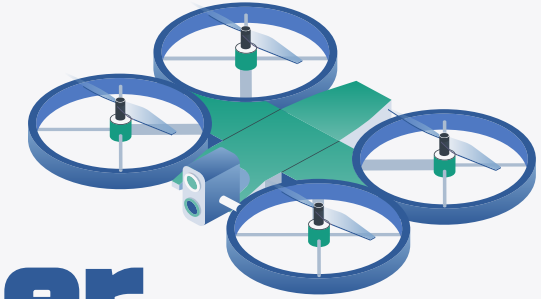
RIT



Spring Semester Report

BAE SYSTEMS

Mason Bausenwein, Dan Corcoran, Aiden Kennedy, Kevin Sakowicz

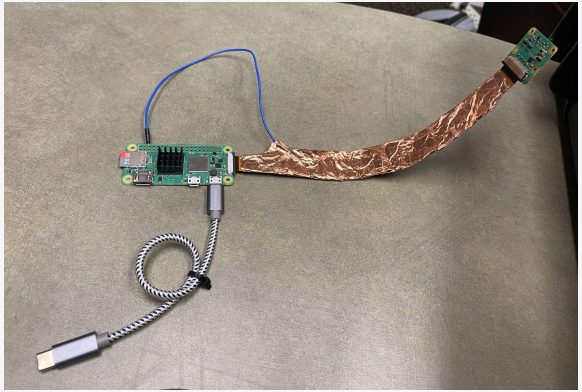
TABLE OF CONTENTS

01 Core Features

02 Architecture

03 Demo

04 Questions

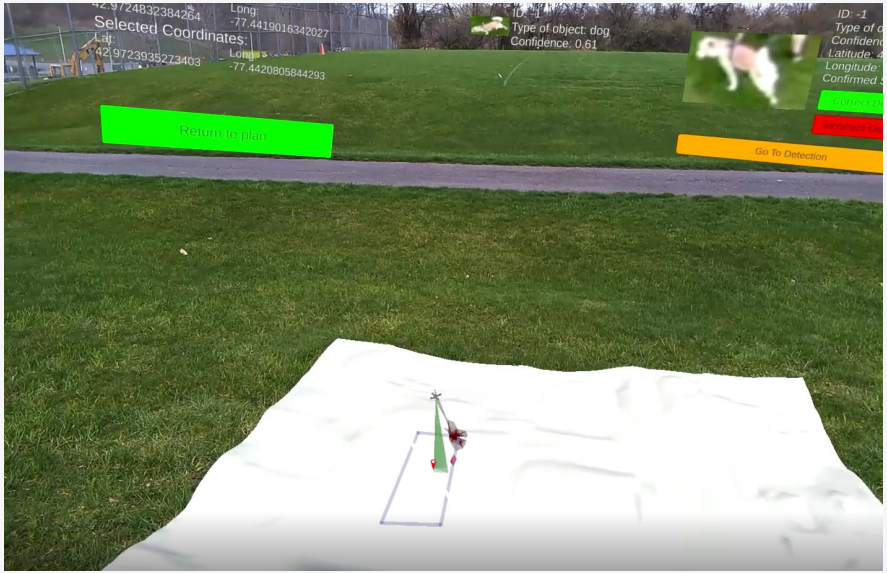


01

Core Features

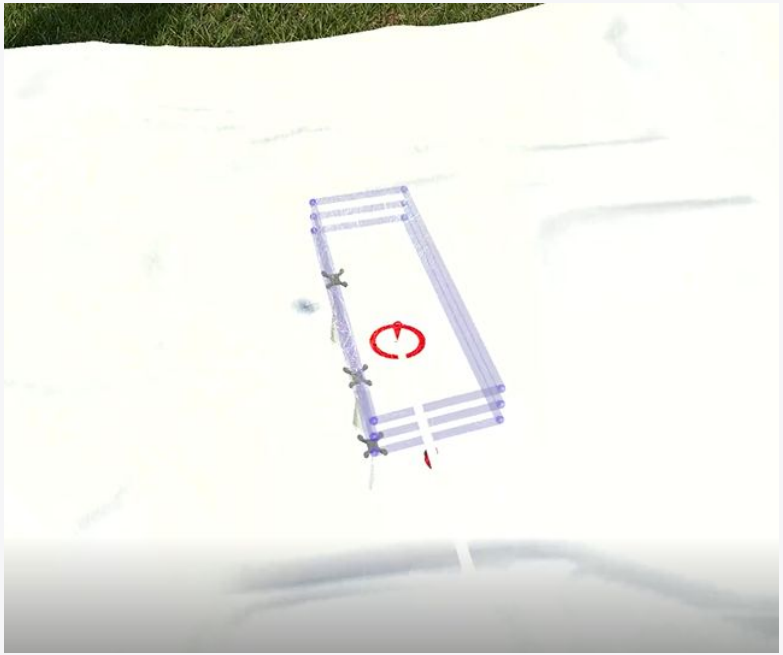
AR Display

- The AR view must show an accurate representation of the terrain from the surveyed location.
- The AR view will track the drones through their flights and display the sent flight plans.
- Drones will update their position above the terrain and display an accurate cone of vision.
- Detections will show in a scroll view in addition to displaying as pins on the terrain.



AR Input

- Operators will be able to select detections to view additional information.
- Operators will be able to manually control the position of drones.
- Operators will be able to confirm or deny detections.
- Multiple operators will be able to use the system simultaneously.



Data Processing Features

Automated Drone Flight

The drones need to fly in a predefined while transmitting their GPS coordinates and yaw, pitch, and roll along with imagery.

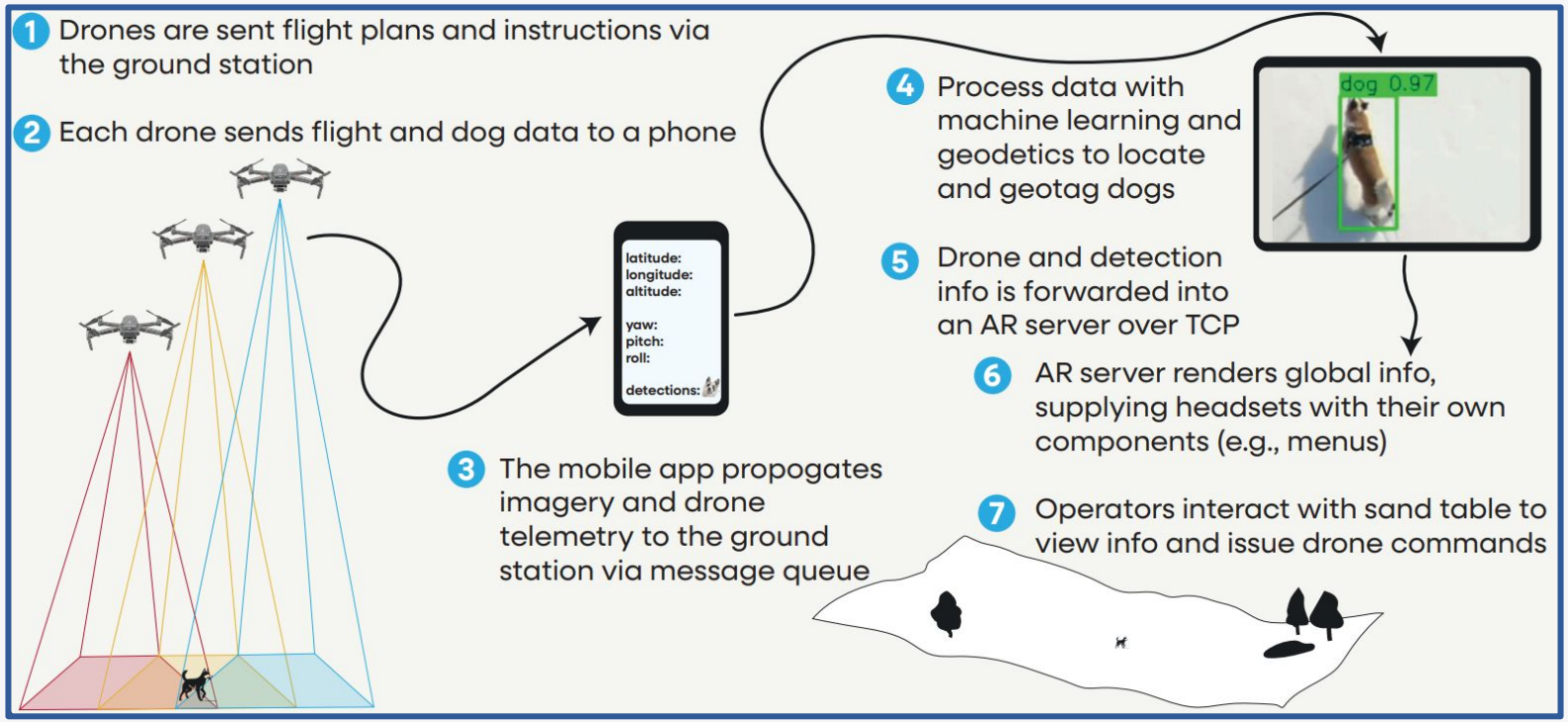
Automated Dog Detection

The system must be able to identify and track dogs from imagery supplied by the drones and forward the location of the dog and an image of it to the AR view.

02

Architecture

Data Flow


1 Drones are sent flight plans and instructions via the ground station
2 Each drone sends flight and dog data to a phone
latitude:
longitude:
altitude:
yaw:
pitch:
roll:
detections:
3 The mobile app propogates imagery and drone telemetry to the ground station via message queue
4 Process data with machine learning and geodetics to locate and geotag dogs
dog 0.97
5 Drone and detection info is forwarded into an AR server over TCP
6 AR server renders global info, supplying headsets with their own components (e.g., menus)
7 Operators interact with sand table to view info and issue drone commands

Trade Offs

Augmented Reality

- Using AR controller for UI rather than hand movements for ease of use.

Android App

- Vendor locked to DJI due to not implementing Mavlink support due to timeconstraints
- Unable to effectively use newer android versions due to the DJI SDK media pipeline not supporting any android version newer than Android 13.

Mission Control

- Java support for machine learning is limited. Writing our machine learning processes in Python gives us more detection capability but requires us to integrate Python processes into a Java pipeline.

Hardware

- Electromagnetic Interference coming from the drones requires us to wrap our CSI cables in copper shielding. This adds rigidity, meaning we cannot use the gimbal with the new cameras.

Future Work

Augmented Reality

- Terrain synchronization anchors for multiple users.
- Live video streaming integration.
- Multi-camera based dynamic terrain.

Mission Control

- Rebroadcast the live video from the Raspberry Pis into a format compatible with Unity.
- Continue training the machine learning model to improve accuracy.
- Continue tuning the tracking algorithm to improve re-identifications

Hardware

- Find proper mounting solution for Raspberry Pis and Pi cameras.
- Create a stable and reconnectable streaming script for the Pis.

03

Demo

04

Questions

Thank You